HireTrack License Admin: Project Overview & Technical Architecture

Mission: To give the vendor a small control plane for commercial HireTrack: who is a paying client, which machine may run the product, and which release that machine is allowed to pull.

The Core Problem and The Solution

HireTrack is a full Node application, not a desktop binary with a license dongle. If you only email a tarball, a customer can copy it to every VM they own. Traditional DRM (online heartbeats every minute, proprietary readers) punishes legitimate operators and still fails against a determined copier.

License Admin is a separate Next.js app (the super-admin site). It does not host the ATS UI. It stores *clients* and *licenses*. A license key is derived from the client email and a machine fingerprint. Validation checks that the key exists, is active, and is bound to that machine. Version listing is a thin proxy over the vendor's GitHub Releases so the installer and in-app updater can see what to download. Revocation is a status flip, not a cryptographic cat-and-mouse game. The installer will not fetch a build until this service accepts the license.

System Architecture & Entitlement Workflow

1. **Client provisioning.** A super-admin signs into License Admin and creates a Client (name, email). No license exists yet. Unknown emails cannot self-register a license; register is rejected until a Client row exists.
2. **First bind on the customer VM.** The installer (or an older `--register` path) collects email plus a machine code and `POST /api/license/register`. The server HMAC-signs `email:machineCode` into a readable license key, stores an *active* License, and binds that machine. A second register for the same email and machine is rejected.
3. **Validate on install and update.** `POST /api/license/validate` receives license key, machine code, and installed version. The server confirms the license is active and the machine matches (or binds the machine if it was still empty). It records `installedVersion` so the dashboard shows what the customer is running.
4. **Release pointer.** On a valid license, the admin returns a GitHub release asset URL for the requested version. That is how installs are gated: no valid license, no tarball URL from this API.
5. **Catalog for the UI.** `GET /api/version/list` and `GET /api/version` read GitHub Releases and return tag names, dates, notes, and download metadata. The ATS settings screen uses this list so operators can pick an update without SSH.
6. **Day-two ops.** Super-admin can read/update clients, revoke or deactivate a license, and download a small license JSON for that client. Revoked licenses fail validate; the customer VM keeps running until the next check, but cannot fetch new builds through this path.

Target Audience Positioning

- **Vendor / super-admin** — the only users of this app. It is not an HR tool.
- **Self-host customers** — they never log into License Admin. Their installer registers and validates against it on every install and update.
- **Release pipeline** — License Admin does not store tarballs. It points a licensed machine at the correct GitHub Release. See the Installer & Versioning Pipeline overview.

Technology Stack Summary

Domain	Technology	Implementation Purpose
Application	Next.js 15, React 19, TypeScript	Super-admin dashboard and JSON APIs in one small service.
Auth	jose / JWT, bcryptjs	Admin login sessions. License keys are not user passwords.
Validation	Zod	Request bodies for register, validate, client CRUD.
Database	MongoDB, Mongoose	Admin, Client, and License collections.
Key derivation	HMAC-SHA256 over email and machine code	Stable, readable license keys bound to one machine.
Release catalog	GitHub Releases HTTP API	Version list and asset URLs; this app does not store the tarball.
Customer edge	HireTrack `installer.sh`	The component on the VM that registers, validates, and then downloads through these APIs.

Primary API surface: `/api/login`, `/api/client/*`, `/api/license/register`, `/api/license/validate`, `/api/license/revoke`, `/api/license/update`, `/api/license/read`, `/api/version`, `/api/version/list`.